

AMD DEVELOPER HACKATHON: ACT II

DescribeX

Create Once. Caption Everywhere.



</> **Open Source** Accessible AI infrastructure.

⚡ **Powered by Fireworks AI** High-performance inference.



DescribeX

AI-Powered Accessible Video Captioning

Create Once. Caption Everywhere.

Upload a short video and instantly generate four AI-powered caption styles with editable exports and burned-in captions.



Upload Video

Drag & drop or click · MP4, MOV, WEBM · up to 100MB · 2 min max

The Problem



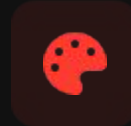
Slow Creation

Generating high-quality captions manually is incredibly time-consuming, bottlenecking the content pipeline.



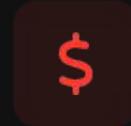
Ecosystem Lock-in

Existing tools often trap creators within closed ecosystems, making it difficult to export or port content.



Rigid Styles

Most AI captioners produce a single, robotic tone, failing to support multiple expressive writing styles.



Expensive

Basic accessibility features should not be locked behind expensive professional software tiers or paywalls.

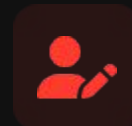
Introducing DescribeX

An open-source, AI-powered video captioning platform built for creators.



AI-Generated Styles

Instantly generate Formal, Sarcastic, and Humorous captions from a single upload.

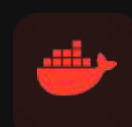


Human-in-the-Loop

AI assists; humans decide. All generated captions are fully editable before final export.



Creator-First Philosophy



Dockerized Engine

Modular **Architecture**



Separation of Concerns

DescribeX consists of two completely independent components: the Next.js Frontend and the standalone Python AI Engine.



RESTful Backend API

The frontend communicates seamlessly with the scalable FastAPI backend to queue and process high-intensity video tasks.



Caption Engine

A specialized pipeline that handles robust video preprocessing, frame extraction, and complex multimodal prompt engineering.



Fireworks AI Integration

Directly utilizes state-of-the-art vision and text language models for rapid, context-aware, and style-specific inference.

Core Features

A premium user experience designed to minimize friction.



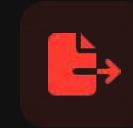
Responsive Modern UI

A clean, Apple-inspired interface with dark mode natively built in. Designed to look beautiful and function flawlessly on any screen.



Multi-Style Generation

Maximize content distribution. Generate four distinct caption variations simultaneously from a single video upload.



Seamless Exporting

Flexibility when you need it. Download generated caption files directly or copy formatted text straight to your clipboard.

Technology Stack



Frontend

Next.js
React 19
TypeScript
Tailwind CSS



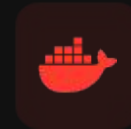
Backend

Python
FastAPI
Uvicorn
Pydantic
Validation



AI Provider

Fireworks
Vision: kimi-k2p6
Text: gpt-oss-120b
Multimodal
Inference



Deployment

Docker
Containerized
Vercel
Engine
Local Execution
(Frontend)

End-to-end execution verified successfully on local development environments.

Containerized Engineering

 Fully compliant with the AMD Developer Hackathon requirements.

Standalone Container

The Python engine is fully dockerized and can run completely independent of the web app.

Input Standard

Mounts to `/input` and automatically reads `tasks.json` for batch processing.

Output Standard

Successfully writes multi-style caption objects to `/output/results.json`.

Deployment **Status**



Verified Successes

Frontend Deployed: Next.js app live on Vercel.

Backend Builds: API starts and connects to Fireworks AI.

Local Execution: Full end-to-end processing works flawlessly on local machines.

Docker Ready: Container runs and processes tasks correctly.



Hosting Constraints

Cloud Resources Pending: Intended for AMD Developer Cloud, but resource approval was pending at deadline.

Temporary Render Limit: Fallback deployment on Render Free Tier is hard-capped at 512MB RAM.

OOM Reality: Video frame extraction predictably exceeds 512MB, causing Out-Of-Memory errors on the live demo server.

Future Roadmap

Advanced Export

Support for direct SRT/VTT downloads and server-side hardcoded video burns.

AMD Acceleration

Transition to AMD Developer Cloud for hardware-accelerated video frame processing.

Authentication & Storage

Implement user accounts and cloud storage to preserve caption history.

Multi-Language

Expand prompt engineering to automatically translate and caption in global languages.

Thank You.

Create Once. Caption Everywhere.

Scan to view the repository

 github.com/Anushiv7/DescribeX

